

IT Security 2

Unit 9 - Blocking known Threats Using Security Profiles

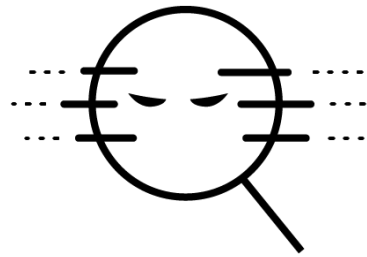
Instructor: Brian Krause

Announcements

Blocking known Threats Using Security Profiles

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

BLOCKING KNOWN THREATS USING SECURITY PROFILES



VERIFY THAT YOUR ALLOWED TRAFFIC IS SAFE

- Security Profile overview
- Vulnerability Protection Security Profiles
- Antivirus Security Profiles
- Anti-Spyware Security Profiles
- File Blocking Profiles
- Data Filtering Profiles
- Attaching Security Profiles to Security policy rules

Learning Objectives

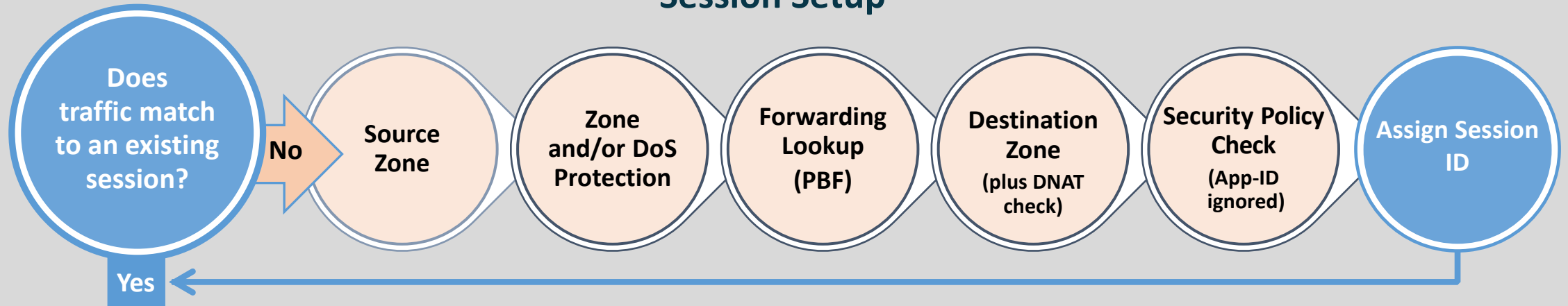
After you complete this module, you should be able to:

- Describe the seven different Security Profile types
- Define the two predefined Vulnerability Protection Profiles
- Configure Security Profiles to prevent virus and spyware infiltration
- Configure File Blocking Profiles to identify and control the flow of file types through the firewall

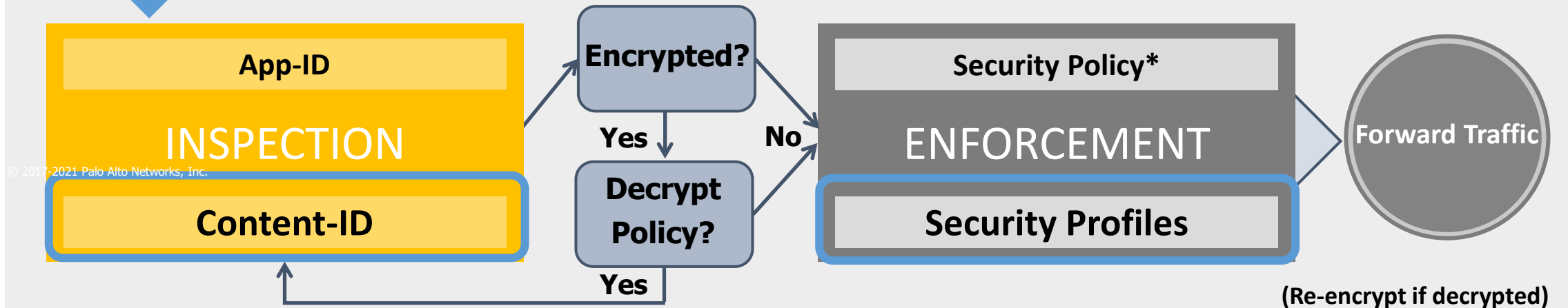


Flow Logic of the Next-Generation Firewall

Session Setup



Inspection and Enforcement



*Policy check relies on pre-NAT IP addresses



Security Profile overview

Vulnerability Protection Security Profiles

Antivirus Security Profiles

Anti-Spyware Security Profiles

File Blocking Profiles

Data Filtering Profiles

Attaching Security Profiles to Security policy rules

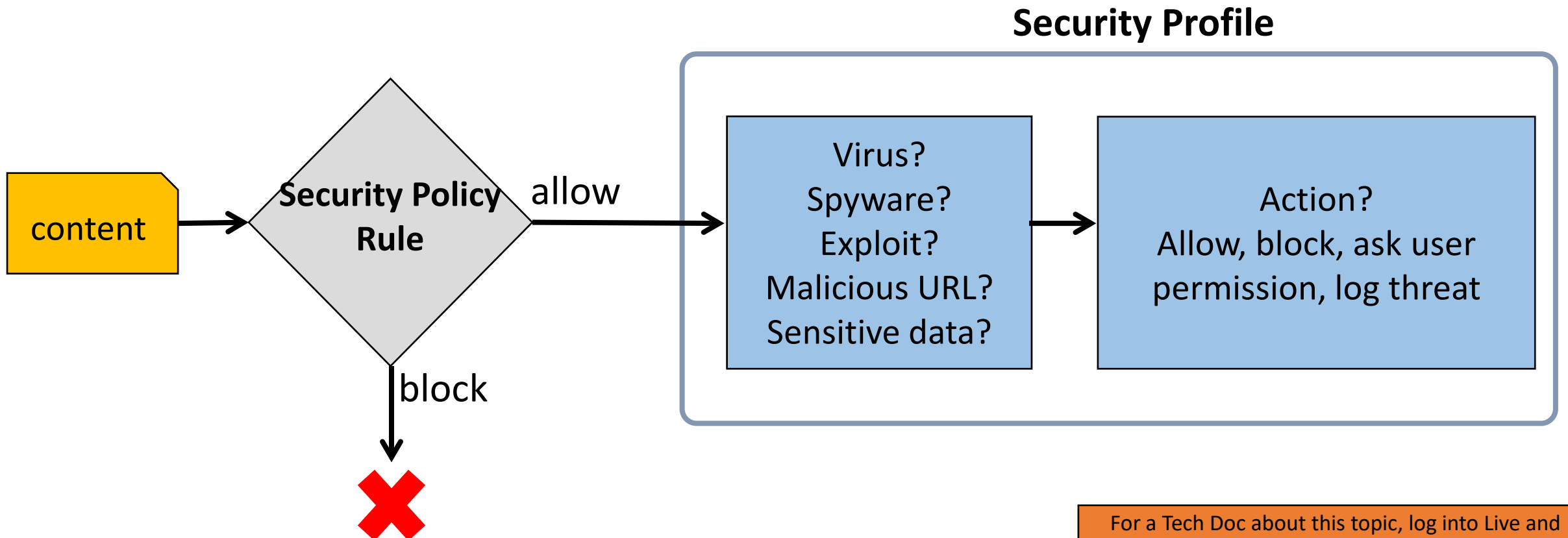
Introducing Content-ID

- The Content-ID feature:
 - Includes an advanced threat prevention engine and policies to inspect and control content traversing the firewall
 - Scans network traffic for:
 - Software vulnerability exploits
 - Viruses
 - Spyware
 - Malicious URLs
 - Restricted files and data

For a Tech Doc about this topic, log into Live and search for "Best Practices for Securing Your Network from Layer 4 and Layer 7 Evasions"

Security Policy with Security Profiles

Security Profiles implement additional security checks on allowed traffic.



For a Tech Doc about this topic, log into Live and search for "Threat Prevention"

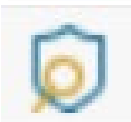
Security Profile Types

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	ACTION	PROFILE
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	application-default	Allow	
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	Internet	any	any	application-default	Allow	
3	Extranet_to_Internet	Extranet	universal	Extranet	any	any	Internet	any	any	application-default	Allow	



Antivirus



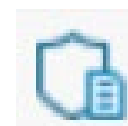
Anti-Spyware



Vulnerability Protection



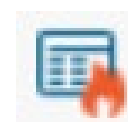
Advanced URL Filtering



File Blocking



Data Filtering



WildFire Analysis



Security Profile Group

For a Tech Doc about this topic, log into Live and search for "Security Profiles"

Threat Log

Vulnerability Protection, Antivirus, and Anti-Spyware Profiles log events to the Threat log.

Monitor > Logs > Threat

		RECEIVE TIME	TYPE	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLI...	ACTION	SEVERITY	URL
		03/22 23:05:28	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
		03/22 23:04:53	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
		03/22 23:04:13	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
		03/22 22:56:03	vulnerability	Terror Exploit Kit Detection	Acquisition	Acquisition	10.3.30.101	159.203.185.4	80	web-browsi...	reset-both	critical	
		03/22 22:55:58	spyware	Trojan.yakes:hellobro.bit	Acquisition	Acquisition	10.10.17.102	104.168.144.17	53	dns	reset-both	medium	hellobro.bit
		03/22 22:55:58	spyware	Trojan.yakes:hellobro.bit	Acquisition	Acquisition	10.10.17.102	10.10.17.1	53	dns	drop-packet	medium	xmr.crypto-pool.f
		03/22 22:55:58	spyware	Trojan.yakes:hellobro.bit	Acquisition	Acquisition	10.10.17.102	104.168.144.17	53	dns	reset-both	medium	hellobro.bit
		03/22 22:55:43	spyware	Trojan.yakes:hellobro.bit	Acquisition	Acquisition	10.10.17.102	104.168.144.17	53	dns	drop	medium	hellobro.bit
		03/22 22:55:38	vulnerability	Non-REC Compliant SMTP	Acquisition	Acquisition	10.11.2.102	66.218.85.52	25	smtp-base	alert	informa...	

Includes threat type

Includes packet capture
(if configured)

Opens Threat Details
window.

For a Tech Doc about this topic, log into Live and search for "Threat Log Fields"

Security Profile overview



Vulnerability Protection Security Profiles

Antivirus Security Profiles

Anti-Spyware Security Profiles

File Blocking Profiles

Data Filtering Profiles

Attaching Security Profiles to Security policy rules

Default Vulnerability Protection Security Profiles

Objects > Security Profiles > Vulnerability Protection

☐	NAME	LOCATION	COUNT	RULE NAME	THREAT NAME	HOST TYPE	SEVERITY	ACTION	PACKET CAPTURE
☐	strict	Predefined	Rules: 10	simple-client-critical	any	client	critical	reset-both	disable
				simple-client-high	any	client	high	reset-both	disable
				simple-client-medium	any	client	medium	reset-both	disable
				simple-client-informational	any	client	informational	default	disable
				simple-client-low	any	client	low	default	disable
				simple-server-critical	any	server	critical	reset-both	disable
				simple-server-high	any	server	high	reset-both	disable
				more...					
☐	default	Predefined	Rules: 6	simple-client-critical	any	client	critical	default	disable
							high	de	
							medium	de	
							critical	de	
							high	default	disable
							medium	default	disable
							medium		

Default profiles

Rules specify actions on detected events.

+

Add

−

Delete

⌙

Clone

📄

PDF/CSV

- To create customized profile actions:
 - **Clone** the default read-only profile and edit the clone, or
 - **Add** a brand-new profile

Vulnerability Protection Profile Rules

Objects > Security Profiles > Vulnerability Protection > Add

The screenshot shows the configuration interface for a Vulnerability Protection Rule. The interface is divided into two main sections: 'Vulnerability Protection Profile' and 'Vulnerability Protection Rule'.

Vulnerability Protection Profile:

- Name: Corp-VP-Profile1
- Description: Corporate VP Profile

Vulnerability Protection Rule:

- Rule Name: Corp-Rule-1
- Threat Name: any
- Action: Default
- Host Type: any
- Packet Capture: disable
- Category: any
- Severity: any (All severities)

Rule Configuration Table:

Any	CVE	VENDOR ID
☐	☒	☒
	CVE-2016-504	MS16-364

Callouts:

- Host Type:** any
- Threat Name:** any
- Action:** Default
- Packet Capture:** disable
- Category:** any
- Severity:** any (All severities)
- Rule Configuration:** Rule will match any signature containing CVE or Vendor ID

Buttons: + Add, - Delete, ↑ Move Up, ↓ Move Down

Vulnerability Exceptions

Objects > Security Profiles > Vulnerability Protection > Add

Vulnerability Protection Profile

Name: Corp-VP-Profile1
Description: Corporate VP Profile

Rules | **Exceptions** | Inline Cloud Analysis

Override the action configured in the rules.

Click to modify packet capture setting.

ENABLE	ID	THREAT NAME	IP ADDRESS EXEMPTIONS	RULE	CVE	HOST	CATEGO...	SEVE...	ACTION	PACKET CAPTURE
☐	92530	Xiaohuanxiong SQL Injection Vulnerability			cve-2022-26268	server	sql-injection	critical	default (reset-server)	disable
☐	92388	Atom CMS Remote Code Execution Vulnerability			cve-2022-25487	server	code-execution	critical	default (reset-server)	disable
☐	92375	ICI Scadaflex II Scada Controller Remote Code Execution Vulnerability			cve-2022-25359	server	code-execution	critical	default (reset-server)	disable
☐	92705	SAP NetWeaver Remote Code Execution Vulnerability			cve-2021-38163	server	code-execution	high	default (reset-server)	disable
☐	92706	SAP NetWeaver Remote Code Execution Vulnerability			cve-2021-38163	server	code-execution	high	default (reset-server)	disable
☐	92717	CODESYS Null Pointer Dereference Vulnerability			cve-2021-34586	server	dos	high	default (reset-server)	disable
☐	92716	CODESYS Null Pointer Dereference Vulnerability			cve-2021-34586	server	dos	high	default (reset-server)	disable

Click to view or add IP addresses.

☒ Show all signatures ☐ PDF/CSV

Page 1 of 609 | Displaying 1 - 30 / 18261 threats

Inline Cloud Analysis

Objects > Security Profiles > Vulnerability Protection > Add

Vulnerability Protection Profile

Name Corp-VP-Profile1

Description Corporate VP Profile

Rules

Exceptions

Inline Cloud Analysis

☒ Enable cloud inline analysis

Available Analysis Engines

MODEL

DESCRIPTION

ACTION

SQL Injection	Detects a common hacking technique where an attacker inserts SQL queries into an applications' request	alert
Command Injection	Detects a common hacking technique that allows an attacker to execute arbitrary operating system (OS) commands on the	alert

Exclude from Inline Cloud Analysis

☐ EDL URL ^

☐ EDL IP ^

+ Add - Delete

+ Add - Delete

SQL and Command Injection analysis engines.

Specify the action to take when an exploit is detected.

Add EDU URL or IP Address to exempt from the profile.

Security Profile overview

Vulnerability Protection Security Profiles

Antivirus Security Profiles



Anti-Spyware Security Profiles

File Blocking Profiles

Data Filtering Profiles

Attaching Security Profiles to Security policy rules

Default Antivirus Security Profile

Objects > Security Profiles > Antivirus

☐	NAME	PACKET CAPTURE	PROTOCOL	Decoders			Application Exceptions		MODEL	WILDFIRE INLINE ML ACTIONS
				SIGNATURE ACTION	WILDFIRE SIGNATURE ACTION	WILDFIRE INLINE ML ACTION	APPLIC...	AC...		
☐	default	☐	http	default (reset-both)	default (reset-both)	default (reset-both)			Windows Executables	
			http2	default (reset-both)	default (reset-both)	default (reset-both)			PowerShell Scripts	
			pop3	default (alert)	default (alert)	default (alert)			PowerShell Scripts	
				default (alert)	default (alert)	default (alert)			Executable Linked Format	
				default (alert)	default (alert)	default (alert)			MSOffice	
			ftp	default (reset-both)	default (reset-both)	default (reset-both)				
			smb	default (reset-both)	default (reset-both)	default (reset-both)				

Predefined profile

Action to take based on antivirus signatures delivered in content updates

WildFire Action to take based on signatures delivered by WildFire

WildFire Inline ML Action to take based on real-time ML scan

- To create customized profile actions:
 - Clone the default read-only profile and edit the clone, or
 - Add a brand-new profile

Creating a New Antivirus Profile

Objects > Security Profiles > Antivirus > Add

Available Actions

default (reset-both)
allow
alert
drop
reset-client
reset-server
reset-both

Antivirus Profile

Name: Corp-AV
Description: Corporate Anti Virus Profile

Action | Signature Exceptions | WildFire Inline ML

☐ Enable Packet Capture

Decoders

PROTOCOL	SIGNATURE ACTION	WILDFIRE SIGNATURE ACTION	WILDFIRE INLINE ML ACTION
smtp	default (alert)	default (alert)	default (alert)
smb	default (reset-both)	default (reset-both)	default (reset-both)
pop3	default (alert)	default (alert)	default (alert)
imap	default (alert)	default (alert)	default (alert)
http2	default (reset-both)	default (reset-both)	default (reset-both)
http	default (reset-both)	default (reset-both)	default (reset-both)
ftp	default (reset-both)	default (reset-both)	default (reset-both)

Application Exceptions

0 items

APPLICATION ACTION

+ Add - Delete

Click to modify to something other than "default" action.

Add applications to exempt from the profile.

Antivirus Profile Signature Exceptions

Objects > Security Profiles > Antivirus > Add

Antivirus Profile ?

Name

Description

Action | Signature Exceptions | WildFire Inline ML

1 item → ×

THREAT ID ^	THREAT NAME	
281328	DOS/Virus.eicar_test_file.	☒

Type a threat ID and click **Add**.

Threat ID

- To reduce the number of false positives, use Threat ID to create an exemption.
- Threat IDs recorded in Threat log

Antivirus Profile WildFire Inline Machine Learning

Objects > Security Profiles > Antivirus > Add

Antivirus Profile

Name

Corp-AV

Description

Corporate Anti Virus Profile

Action

Signature Exceptions

WildFire Inline ML

Available Models

4 items

MODEL	DESCRIPTION	ACTION SETTING
Windows Executables	Machine Learning engine to dynamically identify malicious PE files	disable (for all protocols)
PowerShell Script 1	Machine Learning engine to dynamically detect malicious PowerShell scripts with known length	disable (for all protocols)
PowerShell Script 2	Machine Learning engine to dynamically detect malicious PowerShell scripts without known length	disable (for all protocols)

enable (inherit per-protocol actions)

alert-only (override more strict actions to alert)

disable (for all protocols)

File Exceptions

0 items

☐	PARTIAL HASH	FILENAME	DESCRIPTION
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+ Add

- Delete

For a Tech Doc about this topic,
log into Live and search for
"WildFire Inline ML"

Security Profile overview

Vulnerability Protection Security Profiles

Antivirus Security Profiles

Anti-Spyware Security Profiles



File Blocking Profiles

Data Filtering Profiles

Attaching Security Profiles to Security policy rules

Default Anti-Spyware Security Profiles

Objects > Security Profiles > Anti-Spyware

☐	NAME	COUNT	POLICY NAME	THREAT NAME	SEVERITY	ACTION	PACKET CAPTURE
☐	default	Policies: 4	simple-critical	any	critical	default	disable
			simple-high	any	high	default	disable
			simple-medium	any	medium	default	disable
			simple-low	any	low	default	disable
☐	strict	Policies: 5	simple-critical	any	critical	reset-both	disable
			simple-high	any	high	reset-both	disable
			simple-medium	any	medium	reset-both	disable
			simple-informational	any	informational	default	disable
			simple-low	any	low	default	disable

Predefined profiles

Rules specify actions on detected spyware.

+ Add - Delete Clone PDF/CSV

- To create customized profile actions:
 - Clone the default read-only profile and edit the clone, or
 - Add a brand-new profile

For a Tech Doc about this topic, log into Live and search for "Objects > Security Profiles > Anti-Spyware"

Objects > Security Profiles > Anti-Spyware > Add > Rules

Objects > Security Profiles > Anti-Spyware > Add > Rules



MID-STATE
TECHNICAL COLLEGE

Anti-Spyware Exceptions

Objects > Security Profiles > Anti-Spyware > Add

Anti-Spyware Profile

Name: Coporate-Strict-AS-Profile

Description: Strict coporate anti-spyware profile

Signature Policies | **Signature Exceptions** | DNS Policies | DNS Exceptions

13965 items

ENAB...	ID ^	THREAT NAME	IP ADDRESS EXEMPTIONS	POLICY	CATEGORY	SEVERITY	ACTION	PACKET CAPTURE
☐	10001	Autorun User-Agent Traffic		simple-medium	spyware	medium	default (alert)	disable
☐	10002	Suspicious User-Agent Traffic		simple-medium	spyware	medium	default (alert)	disable
☐	10003	Palevo User-Agent Traffic		simple-medium	spyware	medium	default (alert)	disable
☐	10004	Suspicious User-Agent Strings Detection		simple-low	spyware	low	default (alert)	disable
☐	10005	AVGDM User-Agent Traffic						disable
☐	10006	Small User-Agent Traffic						disable
☐	10007	Fraudload User-Agent Traffic		simple-medium	spyware			disable
☐	10008	Chir User-Agent Traffic		simple-medium	spyware			disable

☒ Show all signatures PDF/CSV

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Can override the action configured in the rules

Click to view or add IP addresses.

Click to override rule's packet capture setting.

Configure DNS Signature Match Protection

Objects > Security Profiles > Anti-Spyware > Add

- Third-party malicious domain lists are made available as EDLs.
- Best practice is to enable sinkhole.
- Can use the Palo Alto Networks IP address or, optionally, your own internal address.

For a Tech Doc about this topic, log into Live and search for "Configure DNS Sinkholing for a List of Custom Domains"

Anti-Spyware Profile

Name: Outbound-AS

Description:

Signature Policies | Signature Exceptions | **DNS Policies** | DNS Exceptions | Inline

DNS Policies

11 items

SIGNATURE SOURCE	LOG SEVERITY	POLICY ACTION	PACKET CAPTURE
External Dynamic Lists			
☐ malicious-domains-edl	medium	sinkhole	disable
Palo Alto Networks Content			
☐ default-paloalto-dns		sinkhole	disable
DNS Security			
☐ Ad Tracking Domains	default		
☐ Command and Control Domains	default		
☐ Dynamic DNS Hosted Domains	default		
☐ Grayware Domains	default		
☐ Malware Domains	default		
☐ Parked Domains	default		

DNS Sinkhole Settings

Sinkhole IPv4: Palo Alto Networks Sinkhole IP (sinkhole.paloaltonetworks.com)

Sinkhole IPv6: IPv6 Loopback IP (::1)

Block DNS Record Types

☐ SVCB ☐ HTTPS ☐ ANY

Custom EDL with malicious domains with policy action configured as sinkhole.

Can use Palo Alto Networks IP or internal IP address

Configure DNS Over HTTPS (DoH)

Objects > Security Profiles > Anti-Spyware > Add

▼ : DNS Security

☐	Ad Tracking Domains	default (informational)	default (allow)	disable
☐	Command and Control Domains	default (high)	default (block)	disable
☐	Dynamic DNS Hosted Domains	default (informational)	default (allow)	disable
☐	Grayware Domains	default (low)	default (block)	disable
☐	Malware Domains	default (medium)	default (block)	disable
☐	Parked Domains	default (informational)	default (allow)	disable

DNS Sinkhole Settings

Sinkhole IPv4

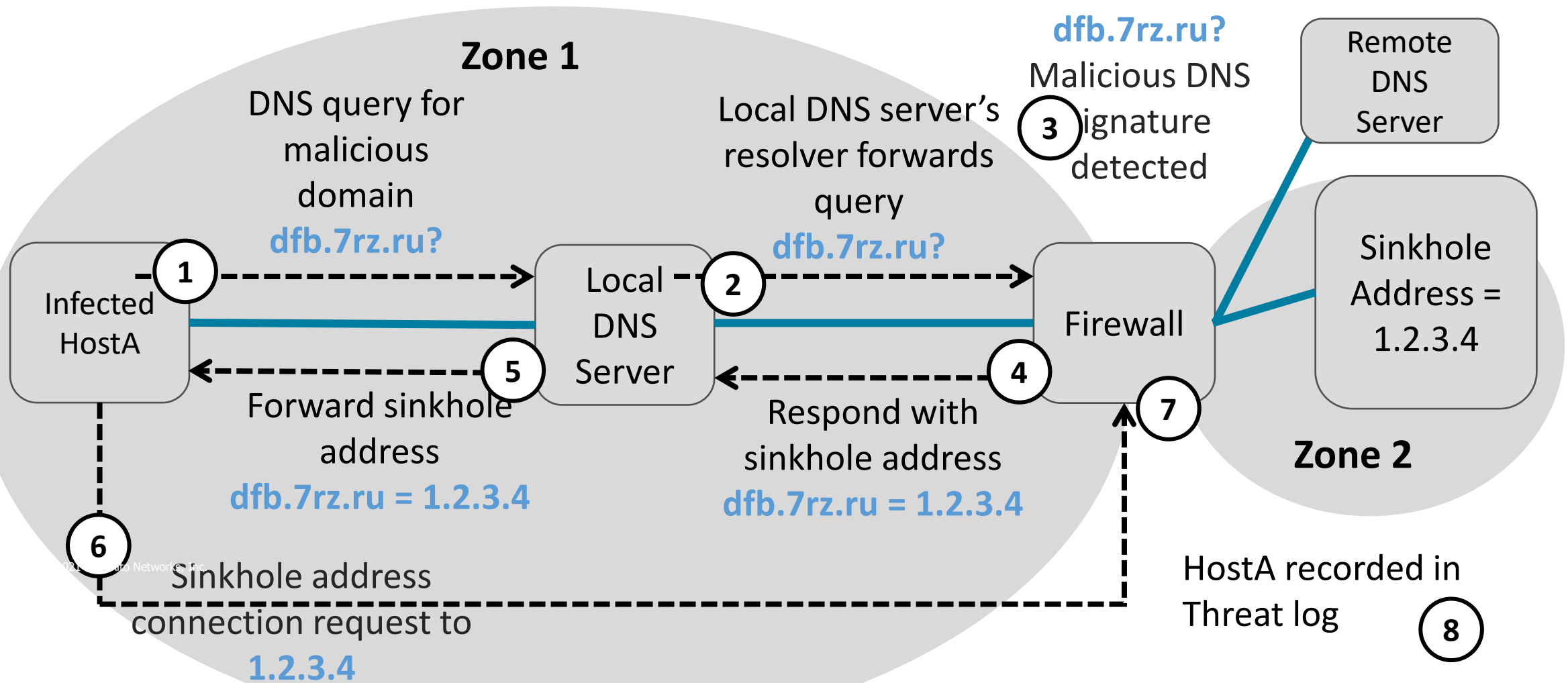
Sinkhole IPv6

Block DNS Record Types

☐ SVCB ☐ HTTPS ☐ ANY

Block encrypted traffic requests to DNS hosts

Sinkhole Operation



DNS Exceptions

Objects > Security Profiles > Anti-Spyware > Add

Anti-Spyware Profile

Name: Outbound-AS

Description:

Signature Policies | Signature Exceptions | DNS Policies | **DNS Exceptions** | Inline Cloud Analysis

DNS Domain/FQDN Allow List

☐	DOMAIN/FQDN	DESCRIPTION
☐	example.com	Example domain

+ Add - Delete

DNS Signature Exceptions

0 items → ×

ENABLE	THREAT ID ^	DOMAIN/FQDN	THREAT NAME
Search for DNS Signature exception(s) and enable			

Add Domain Name or FQDN

Add specific Threat IDs to exclude from policy enforcement

Inline Cloud Analysis

Objects > Security Profiles > Anti-Spyware > Add

Anti-Spyware Profile

Name: Outbound-AS

Description:

Signature Policies | Signature Exceptions | DNS Policies | DNS Exceptions | **Inline Cloud Analysis**

☒ Enable cloud inline analysis

Available Analysis Engines

MODEL		ACTION
HTTP Command and Control detector	Machine Learning engine to detect HTTP based command and control traffic	alert
HTTP2 Command and Control detector	Machine Learning engine to detect HTTP2 based command and control traffic	alert
SSL Command and Control detector	Machine Learning engine to detect SSL based command and control traffic	alert
Unknown-TCP Command and Control detector	Machine Learning engine to detect Unknown-TCP based command and control traffic	alert
Unknown-UDP Command and Control detector	Machine Learning engine to detect Unknown-UDP based command and control traffic	alert

Exclude from Inline Cloud Analysis

☐ EDL URL ☐ IP ADDRESS

+ Add - Delete

Cloud Inline Analysis Engines.

Add EDU URL or IP Address to exempt from the profile.

View Malicious Domains in the Threat Log

Monitor > Logs > Threat

	RECEIVE TIME	TYPE	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLI...	ACTION	SEVERITY	URL
	03/22 23:05:28	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
	03/22 23:04:53	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
	03/22 23:04:13	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com
	03/22 22:54:38	spyware	malicious-domains-edl	Users_Net	Internet	192.168.1.20	8.8.8.8	53	dns	sinkhole	medium	www.quora.com

Internal DNS server or infected host

Typically, an external DNS server

- If you see a sinkhole in the Threat log:
 - Filter the Traffic log to see who is attempting to connect to the sinkhole IP address.
 - Hosts attempting to connect to the sinkhole address could be infected.
- Check logs daily or run a daily report.

For a Tech Doc about this topic, log into Live and search for "Threat Log Fields"

Security Profile overview

Vulnerability Protection Security Profiles

Antivirus Security Profiles

Anti-Spyware Security Profiles

File Blocking Profiles

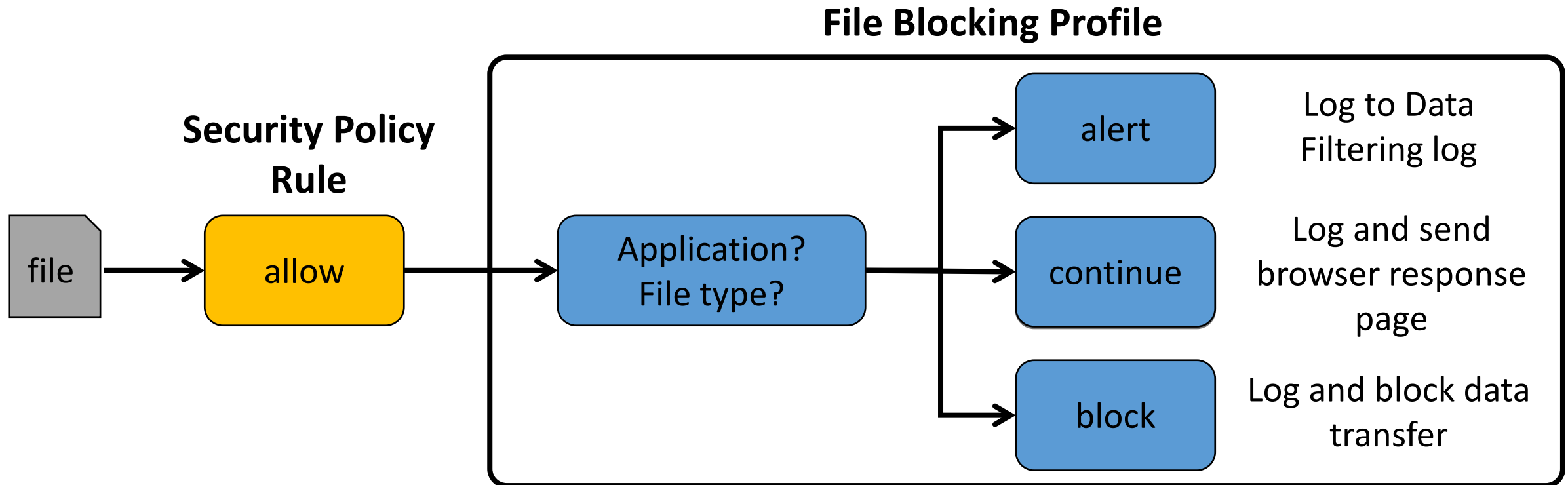


Data Filtering Profiles

Attaching Security Profiles to Security policy rules

File Blocking Overview

- Prevent introduction of malicious data
- Prevent exfiltration of sensitive data
- Logs to Data Filtering log



Creating a New File Blocking Profile

Objects > Security Profiles > File Blocking > Add

File Blocking Profile

Name

Corporate-FB-Profile

Description

Threat prevention through blocking files types

2 items

☐	NAME	APPLICATIONS	FILE TYPES	DIRECTION	ACTION
☐	Rule 1	web-browsing	any	both	alert
☐	Rule 2	any	any	both	continue

7z

access-shortcut

ace

ade

adp

ai

aip-encrypted-d...

aip-encrypted-p...

aip-encrypted-x...

apk

arj

upload

download

both

alert

block

continue

+

Add

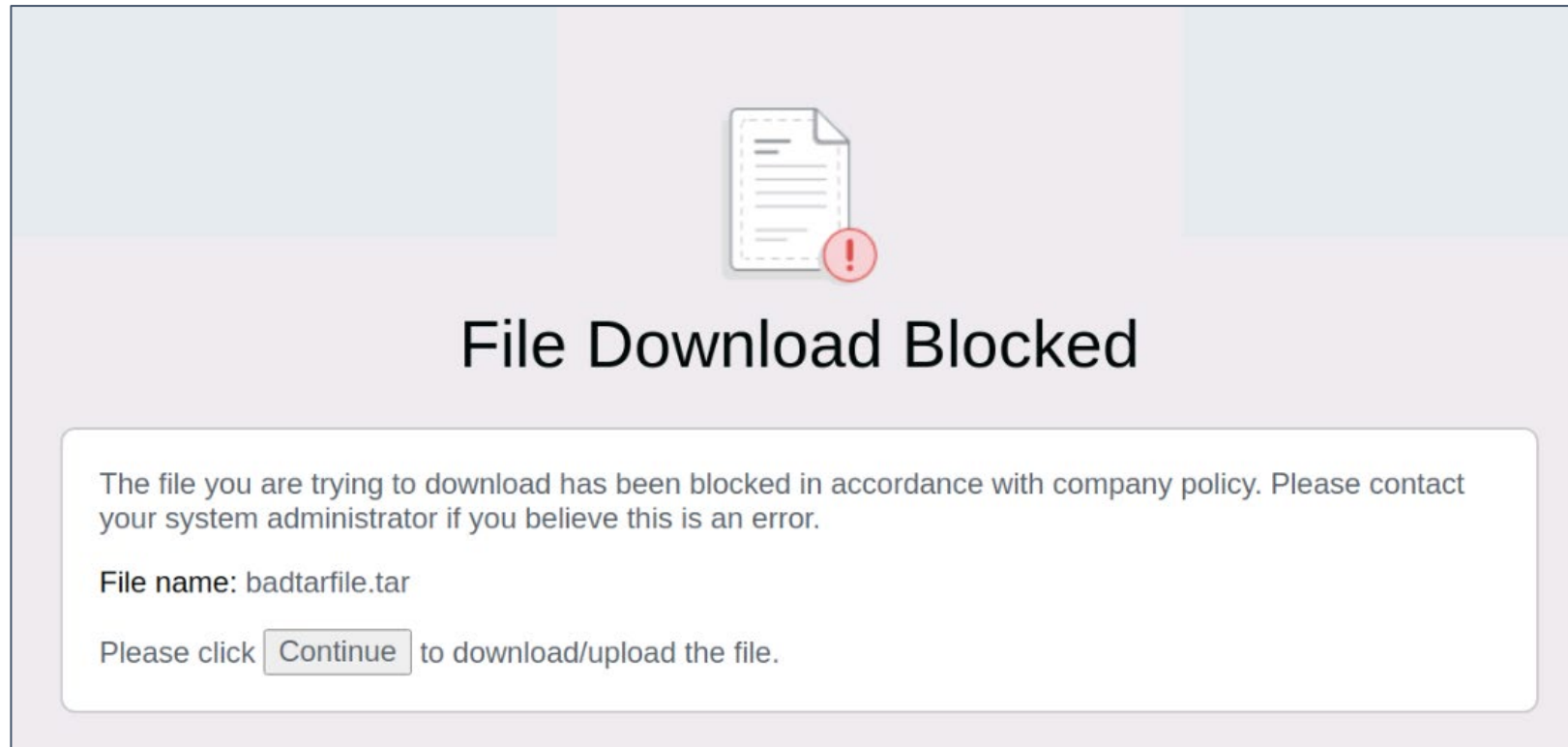
-

Delete

Add one or more rules to control file transfer.

Continue Response Page

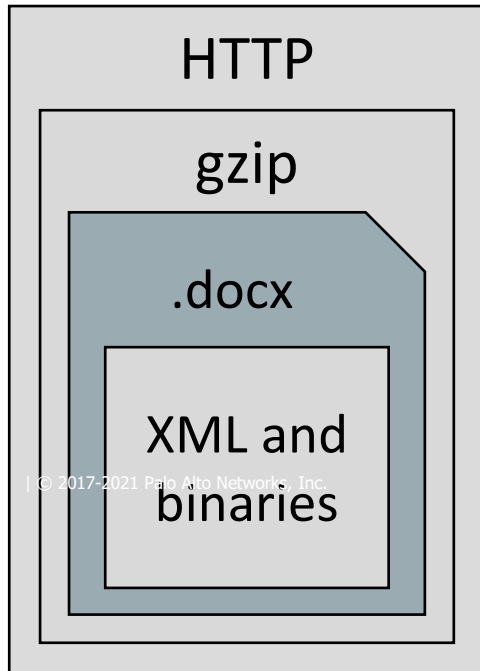
- A “continue” action requires user permission to complete the file transfer.
- Operates only when paired with the application web-browsing



Blocking Multi-Level Encoded Files

Objects > Security Profiles > File Blocking > Add

Firewall decodes
max of four levels



File Blocking Profile

Name: Block-Multi-Level-Encoded-Files

Description: Block files that are multi-level encoded

1 item

☐	NAME	APPLICATIONS	FILE TYPES	DIRECTION	ACTION ^
☐	Block-Multiple-Levels	any	Multi-Level-Encoding	both	block

+ Add - Delete

Blocks files encoded more than four levels.

View Blocked Files in the Data Filtering Log

- Data Filtering log records the file name and file type for blocked files.
- Source is the system that sent the file.
- Destination is the system that received the file.

Monitor > Logs > Data Filtering

		RECEIVE TIME	CAT...	FILE NAME	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLICATION	ACTION
		03/22 22:56:03	any	gwccax.exe-rc4.exe	Unknown Binary File	Acquisition	Acquisition	10.3.30.101	159.203.185.4	80	web-browsing	alert
		03/22 22:56:03	any	SilverApp1.dll	Microsoft PE File	Acquisition	Acquisition	10.3.30.101	159.203.185.4	80	silverlight	deny
		03/22 22:56:03	any	SilverApp1.xap	ZIP	Acquisition	Acquisition	10.3.30.101	159.203.185.4	80	silverlight	alert
		03/22 22:55:53	any	gate.php	Hypertext Preprocessor PHP File	Acquisition	Acquisition	10.10.17.102	188.209.52.233	80	web-browsing	alert
		03/22 22:55:43	any		US-SSNs	Acquisition	Acquisition	10.10.17.102	204.79.197.200	80	web-browsing	reset-both
		03/22 22:55:38	any		Email Link	Acquisition	Acquisition	10.11.2.102	66.218.85.52	25	smtp-base	alert
		03/22 22:55:38	any	logout.php	Hypertext Preprocessor PHP File	Acquisition	Acquisition	10.11.2.102	118.193.174.133	80	web-browsing	alert
		03/22 22:55:33	any		Email Link	Acquisition	Acquisition	10.11.2.102	74.125.24.27	25	smtp-base	alert
		03/22 22:55:23	any	logout.php	Hypertext Preprocessor PHP File	Acquisition	Acquisition	10.11.2.102	118.193.174.133	80	web-browsing	alert
		03/22 22:55:18	any		Email Link	Acquisition	Acquisition	10.11.2.102	66.218.85.139	25	smtp-base	alert
		03/22 22:55:08	any	logout.php	Hypertext Preprocessor PHP File	Acquisition	Acquisition	10.11.2.102	118.193.174.133	80	web-browsing	alert

Security Profile overview

Vulnerability Protection Security Profiles

Antivirus Security Profiles

Anti-Spyware Security Profiles

File Blocking Profiles

Data Filtering Profiles



Attaching Security Profiles to Security policy rules

Creating a Data Pattern

For a Tech Doc about this topic, log into Live and search for "Predefined Data Filtering Patterns"

Objects > Custom Objects > Data Patterns > Add

Data Patterns

Name

Confidential Data

Description

Restricted Corporate and Personal Data

Pattern Type

Predefined Pattern

3 items

☐	NAME	DESCRIPTION	FILE TYPE
☐	Credit Card Numbers	US Credit Card Numbers pattern	Any
☐	Social Security Numbers	US Social Security Numbers pattern	Any
☐	Social Security Numbers (without dash separator)	US Social Security Numbers pattern without dash	Any

Predefined Pattern

Regular Expression

File Properties

+ Add

- Delete

Creating a Data Filtering Profile

For a Tech Doc about this topic, log into Live and search for "Create a Data Filtering Profile"

Objects > Security Profiles > Data Filtering > Add

Data Filtering Profile

NameBlock-Corp-Confidential-Data

DescriptionBlocking CC and SS numbers

☐ Data Capture

1 item

	DATA PATTERN	APPLICATIONS	FILE TYPE	DIRECTION	ALERT THRESHOLD	BLOCK THRESHOLD	LOG SEVERITY
☐	Confidential Data	any	Any	both	0	0	informational

upload

download

both

+ Add

- Delete

Alert/Block Threshold values: (0-65535)

Number of times data pattern must be detected before alert

Number of data pattern instances

View the Data Filtering Log

- Data Filtering log records the file name and file type.
- Source is the system that sent the file.
- Destination is the system that received the file.

Monitor > Logs > Data Filtering

		RECEIVE TIME	CAT...	FILE NAME	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLICATION	ACTION
		03/12 21:31:01	any	AMJjTK02zjzkwP0ioN-EWDO	Google Chrome Extension CRX File	Users_Net	Internet	192.168.1.20	142.250.34.2	80	web-browsing	alert
		03/12 18:03:51	any	apacheselfsigned.pfx	DER Encoded X509 Certificate	Extranet	Users_Net	192.168.50.80	192.168.1.20	47145	ftp	alert
		03/12 18:03:51	any	apache-selfsigned.crt	PEM Encoded X509 Certificate	Extranet	Users_Net	192.168.50.80	192.168.1.20	56967	ftp	alert
		03/12 18:01:44	any	AO4b1Vv0blkrRVGIP54MKrI	Google Chrome Extension CRX File	Users_Net	Internet	192.168.1.20	142.250.34.2	80	web-browsing	alert
		03/12 18:01:43	any	XZyR5pQ7gb8FpWW3-n-P3Q	Google Chrome Extension CRX File	Users_Net	Internet	192.168.1.20	173.194.55.108	80	web-browsing	alert

Security Profile overview

Vulnerability Protection Security Profiles

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Attaching Security Profiles to Security policy rules



Assigning Security Profiles to Security Rules

Policies > Security > Add

The screenshot displays the 'Security Policy Rule' configuration window. The 'Actions' tab is selected, showing the 'Action Setting' section with 'Action' set to 'Allow' and an unchecked 'Send ICMP Unreachable' option. Below this is the 'Profile Setting' section, which contains a 'Profile Type' dropdown set to 'Profiles' and a 'Group Profile' dropdown set to 'Corporate-Security-Profile-Grp'. An arrow points from the 'Profiles' option in the 'Profile Type' dropdown to the 'Group Profile' dropdown. Below the 'Profile Setting' section, several other security features are listed with their respective profiles: Antivirus (Strict-Corporate-Antivir), Vulnerability Protection (Strict-Corporate-Vulnerability), Anti-Spyware (Strict-Corporate-Anti-Spyware), URL Filtering (Strict-Corporate-URL-Filter), File Blocking (Strict-Corporate-File-Block), Data Filtering (Strict-Corporate-Data-Filter), and WildFire Analysis (None).

- Assign individual Security Profiles to a Security policy rule, or
- Assign a Security Profile Group to a Security policy rule

For a Tech Doc about this topic, log into Live and search for "Create a Security Profile Group"

Security Profile Groups

Objects > Security Profile Groups > Add

Security Profile Group	
Name	Corporate-Security-Profile-Grp
Antivirus Profile	Strict-Corporate-Antivirus
Anti-Spyware Profile	Strict-Corporate-Anti-Spyware
Vulnerability Protection Profile	Strict-Corporate-Vulnerability
URL Filtering Profile	Strict-Corporate-URL-Filter
File Blocking Profile	Strict-Corporate-File-Block
Data Filtering Profile	Strict-Corporate-Data-Filter
WildFire Analysis Profile	None

- Add Security Profiles that are commonly used together
- Security Profile Groups simplify Security policy rule administration.

Security Policy Rules

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	ACTION	PROFILE
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	application-default	Allow	
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	Internet	any	any	application-default	Allow	
3	Extranet_to_Internet	Extranet	universal	Extranet	any	any	Internet	any	any	application-default	Allow	

Hover mouse pointer over profile icon displays profile name

Antivirus Profiles: Strict-Corporate-Antivirus

	NAME	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE
			ZONE	ADDRESS	USER	ZONE	ADDRESS					
1	Users_to_Extranet	universal	Users_Net	any	any	Extranet	any	any	application-default	any	Allow	
2	Users_to_Internet	universal	Users_Net	any	any	Internet	any	any	application-default	any	Allow	
3	Extranet_to_Internet	universal	Extranet	any	any	Internet	any	any	application-default	any	Allow	

Hover mouse pointer over group icon displays group name

Profile Group: Corporate-Security-Profile-Grp

Module Summary

Now that you have completed this module, you should be able to:

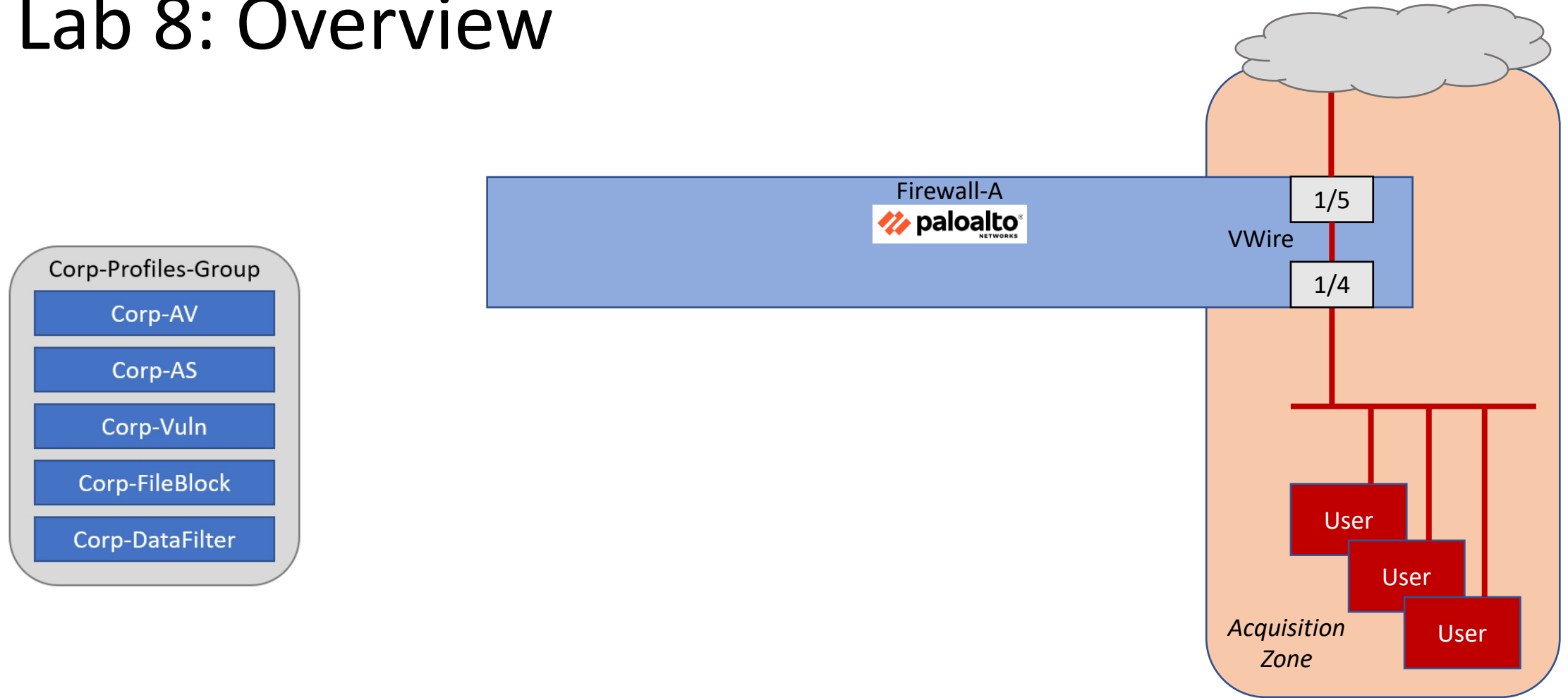
- Describe the seven different Security Profile types
- Define the two predefined Vulnerability Protection Profiles
- Configure Security Profiles to prevent virus and spyware infiltration
- Configure File Blocking Profiles to identify and control the flow of file types through the firewall



Questions



Lab 8: Overview



Lab 8: Blocking Known Threats Using Security Profiles

- Load a baseline configuration
- Generate traffic without profiles and examine logs
- Create Security Profiles
- Create a Security Group
- Apply the Security Group to existing Security policy rules
- Generate traffic with profiles and examine logs

End Content

Unit 9 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 9 Lab: Netlabs Lab 8 – Blocking known Threats Using Security Profiles
- Unit 9 Quiz